
CAPACITY CLIFF AND CALIBRATION ABLATION IN A HYBRID QUANTUM--CLASSICAL CLASSIFIER

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Abstract

This work is a controlled examination of using a quantum circuit as the output layer of a classifier, not an attempt to demonstrate quantum advantage. Building on the hybrid Bayesian quantum--classical classifier of Wang et al. (2026), we reproduce its 4-qubit quantum layer with CUDA-Q and extend it with two experiments that the original study does not cover: (i) a **capacity cliff** scan over qubit count and circuit depth, testing whether performance is monotonic in trainable capacity; and (ii) a **calibration ablation** in which a dephasing operator makes the quantum layer classical, isolating the marginal contribution of quantumness to calibration. Every quantum-circuit result is computed on both CUDA-Q and an independently implemented pure-NumPy state-vector simulator, then compared: the reproduction pipeline agrees to 4.16×10^{-17} , the largest capacity-scan circuit (250 gates) to 3.47×10^{-18} , and the ablation circuit to 8.33×10^{-17} . We explicitly do not claim any quantum advantage: the state vector of 5 qubits occupies 512 bytes and is fully classically simulable. The contribution is a fully verifiable pipeline and a falsifiable experimental design.

Keywords quantum Bayesian networks · hybrid quantum--classical learning · calibration · uncertainty quantification · CUDA-Q

1 Introduction

The final layer of a deep-learning classifier usually outputs a **softmax** distribution—a set of real numbers exponentiated and normalised so that they sum to 1, and therefore probability-shaped. But these numbers are **not** calibrated probabilities: they are frequently and systematically overconfident [1]. By **calibration** we mean whether the confidence a model claims matches how often it is actually right—a model that claims 90% confidence but is right only 70% of the time cannot be trusted even if its accuracy is high. In domains where trustworthy confidence is required (medicine, finance, education) this problem cannot be assessed by accuracy alone.

A class of hybrid architectures has appeared in recent years: a classical network extracts features and a quantum circuit then acts as the decision layer [2]. Such work reports improvements in both accuracy and calibration, but it shares one structural feature—the **quantum layer is never treated as a controlled variable in an ablation**. Take [2]: its state-of-the-art design is a “single-variable” comparison in which all models share the same preprocessing and classification head, and only the first convolutional layer is changed from deterministic to Bayesian. The quantum layer is an identical constant across both arms.

Consequently that study—and work of the same kind—**cannot answer whether quantumness itself is necessary**. This work supplies that missing step.

Contributions

1. We propose a **dephasing ablation**: using Tucci’s classicalising operator [3] as the sole variable **within a single model** (no change of model, data or parameter count) to examine the marginal contribution of quantumness to calibration and capacity. Within the scope of our search we found no prior work that uses this operator as an ablation tool—the closest efforts either attribute behaviour observationally via entanglement measures [4], or introduce depolarising noise as an external variable [5].
2. We extend the **capacity cliff** scan to qubit counts $3\text{--}10 \times$ circuit depths $1\text{--}8$. Prior work has established the (Q, L) scaling protocol and reported saturation [6], and has observed non-monotonicity along a single axis ([7] for the qubit axis; [8] for the depth axis). What we add is the complete two-dimensional grid under a **fixed readout rule, with calibration reported alongside accuracy**, and one phenomenon that prior work has not reported: **test accuracy decreasing with qubit count**—which holds for configurations in which capacity is genuinely used (the per- n peak and the deepest slice are monotone from $n = 4$), while shallower slices are masked by under-optimisation. ($n \geq 3$ is a hard requirement: eight class probabilities need at least a 3-qubit projective readout.)
3. We reproduce the 4-qubit quantum layer of [2] with CUDA-Q and cross-validate it against an independent NumPy implementation (4.16×10^{-17}); this item is positioned as **reproducibility evidence**, not as a methodological contribution. The two experiments of Section 6 are endorsed separately on the reference track: the largest capacity-scan circuit at 3.47×10^{-18} and the ablation circuit at 8.33×10^{-17} .
4. We honestly report three negative conclusions (see Section 6).

2 Related Work

2.1 Hybrid quantum–classical classifiers

The three-stage skeleton of [2] is “classical feature extraction $\rightarrow$ quantum state evolution $\rightarrow$ classical decision”. Its quantum layer uses a 4-qubit, depth-2 parameterised quantum circuit (PQC), and the readout measures the Pauli- Z expectation values of all qubits into a vector that a linear layer then classifies. Its gains come explicitly from the **classical Bayesian front end**: MNIST (a handwritten-digit image dataset) +2.32 and Fashion-MNIST (a clothing-image dataset) +5.61 percentage points, with the quantum layer held fixed.

2.2 Known limits of capacity and distillation

[9] reports two related phenomena in the study of distillation for static word embeddings. First, when the student dimension is reduced to $d = 64$, knowledge distillation actually degrades performance ($63.8 \rightarrow 52.5$). Second, switching to a stronger teacher (GTE-large, a general-purpose text-embedding model) makes the student worse (62.9 against 63.8 for GTE-base), which the authors attribute to an excessive teacher–student capacity gap. This work tests whether the same phenomenon—failure once the capacity gap becomes too large—also appears in a quantum decision layer.

2.3 Quantum Bayesian networks

[3] generalises the nodes of a classical Bayesian network from conditional probability tables to quantum channels, and shows that applying the dephasing operator cl to a node causes the entire network to degenerate into a classical Bayesian network. This operator is the theoretical basis of our ablation experiment: it is an **exact classicalisation map**, so “quantum versus classical” requires no change of model, data or parameter count.

2.4 Quantumness as a controlled variable

The three works closest to ours each take a different route, and each leaves the gap this study fills.

[10] replaces softmax with a **complex-valued unitary representation** as the classification head: one shared backbone, only the lightweight head swapped. On CIFAR-10 (a ten-class colour-image dataset) it lowers the expected calibration error (ECE) from 0.0355 to 0.0146, yet in the same study replacing the readout with a Born-rule measurement layer (a readout that takes the squared amplitude of the wave function as the probability) **degrades** it to 0.0819. Two things follow: single-variable ablation of this kind is feasible, and a **quantum-style readout is not by itself a guarantee of calibration**. Note, however, that their variable is the **form of the representation** (complex unitary versus real), and their unitary layer is parameterised by the Cayley map (a way of writing a unitary matrix in terms of an anti-Hermitian one) and therefore exactly executable on classical hardware—so what they isolate is still not quantumness.

[4] attributes performance **observationally** through the Meyer–Wallach entanglement measure (a way of quantifying how entangled a many-body state is) on a paraphrase-detection task—deciding whether two sentences mean the same thing ($r = 0.85$). But correlation is not intervention: entanglement co-varies with circuit depth and parameter count, so confounding cannot be excluded.

[5] instead uses **depolarising noise** as an external variable, sweeping the ECE of variational classifiers, finds that ECE does not rise monotonically with noise, and warns that a less expressive ansatz may look well calibrated merely because it has collapsed to a degenerate solution.

The shared gap is that quantumness itself has never been the sole variable **within a single model**. This study fills that position with the classicalising operator of [3]. Taking up the warning of [5], we record a collapse indicator for every (arm, seed) pair.

3 Theoretical Background

This section states, as provable propositions, why the dephasing ablation works and at which circuit positions it can work at all—using **only** tools covered by the two designated university textbooks: Griffiths and Schroeter (hereafter **G**) and Arfken, Weber and Harris (**A**). The required background is undergraduate quantum mechanics: density operators, projection operators, unitary transformations and commutators.

Tool used below	Source	Where it is used
Density operator, trace and purity	G §12.3	3.1 pure vs. mixed states
Measurement probability = diagonal, $P(i) = \rho_{ii}$	G §12.3.1 Eq. 12.16	Most critical: proof of Theorem 2 in 3.4
Projection operators and completeness $\sum_n P_n = I$	G §3.6.2 Eq. 3.91, 3.93	Definition in 3.2, Lemma 1 in 3.3
Evolution of the density operator	G §12.3.2 Eq. 12.26	3.1 why the word “channel” is needed
Pauli matrices and spin rotations	G §4.4	the R_Y counterexample in 3.3
Commutator $[A, B] = AB - BA$	A §5.3 Eq. 5.42	Most critical: Theorem 1 in 3.3
Functions of operators and Baker–Hausdorff	A §2.2 Eq. 2.85	3.3 criterion for commutation
Euler identity $\exp(i\sigma_k\theta)$	A §2.2 Eq. 2.80	3.3 closed form of R_Y
Unitary similarity transformation	A §5.6 Eq. 5.76	3.3 proof of Theorem 1
Simultaneous diagonalisation $\leftrightarrow$ commutation	A §6.4	3.3 equivalent statement
Sequential measurements: why order matters	GS Problem 3.33	3.5 minimal counterexample

Table 1: Textbook map for the theoretical section. G = Griffiths 3e, A = Arfken 7e, GS = Griffiths solutions manual. Every row is original content in those books and can be looked up directly.

The one thing the textbooks do not contain, and why that is not an obstacle. The two **names** “quantum channel” and “Kraus operator” do not appear in G or A: a verbatim search for Kraus, channel, quantum operation, superoperator, POVM and completely positive returns **zero** hits in both books. But what this section actually **uses** is the single operation “knock out the off-diagonal entries of a density matrix with a set of projection operators”, and projection operators with completeness are original content in G §3.6.2 (Eq. 3.91, 3.93). In other words, Section 3.1 introduces the channel vocabulary only so that the presentation matches the modern literature; **every deduction below can be restated entirely in the language of G and A**. If the general theory is questioned, the external reference is Nielsen and Chuang [11], and nothing else in this section depends on it.

3.1 Density operators and quantum channels

Pure and mixed states. The state of a quantum system is described by a **density operator** ρ : a positive semi-definite operator with trace 1. If $\rho^2 = \rho$ (equivalently $\text{Tr } \rho^2 = 1$) then $\rho = |\psi\rangle\langle\psi|$ is a **pure state**; otherwise it is a **mixed state**. A pure state encodes maximal knowledge of the system; a mixed state encodes ignorance.

Why mixed states are unavoidable here. Two situations in this work require them: (i) after dephasing is applied to a quantum state the result is no longer pure; and (ii) noise on real hardware also makes states mixed. Neither can be expressed with a state vector $|\psi\rangle$ alone.

Quantum channels. Once a system couples to its environment, or is measured, its evolution is no longer unitary. The general tool for such evolution is a **quantum channel**: a linear map $\mathcal{E}$ taking density operators to density operators that is (i) **trace-preserving**: $\text{Tr } \mathcal{E}(\rho) = \text{Tr } \rho$; and (ii) **completely positive**: applying $\mathcal{E}$ to only part of a larger system still yields a legitimate density operator—no negative probabilities appear. The second condition is what distinguishes a genuine quantum channel from an arbitrary positive map, and is why it is physically realisable. Together the two conditions define a **CPTP map** (completely positive trace-preserving map), the standard definition of a physically implementable quantum operation [11].

The Kraus representation theorem. Every CPTP map can be written as

$$\mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger, \quad \sum_i K_i^\dagger K_i = I,$$

where the K_i are the **Kraus operators** of the channel. The representation is not unique, but its existence is equivalent to being CPTP. Nothing below uses the technical details of complete positivity beyond this representation.

3.2 Computational-basis dephasing is classicalisation

Definition (computational-basis dephasing). For a system of n qubits let $P_b = |b\rangle\langle b|$ be the rank-one projection onto a computational basis state (b ranges over all n -bit strings, and $\sum_b P_b = I$). Define

$$\mathcal{E}(\rho) = \sum_b P_b \rho P_b.$$

Reading this entrywise: multiplying on the left by P_b keeps only row b , and multiplying on the right by P_b keeps only column b ; after summing, only the diagonal survives:

$$\mathcal{E}(\rho) = \text{diag}(\rho),$$

that is, **every off-diagonal entry is set to zero**. This is a CPTP map whose Kraus operators are simply $K_b = P_b$, since $\sum_b P_b^\dagger P_b = \sum_b P_b = I$.

Physical meaning. An off-diagonal entry ρ_{ij} with $i \neq j$ encodes the **phase relation** between basis states $|i\rangle$ and $|j\rangle$ —the source of quantum interference. Zeroing the off-diagonal entries therefore **removes the system’s entire capacity to interfere**, and its behaviour degenerates to a classical probability distribution. This is exactly the classicalising operator cl of Tucci [3]: applying cl to the nodes of a quantum Bayesian network collapses the whole network to a classical Bayesian network.

Why this makes an ideal ablation tool. An ablation can only claim to isolate a single variable if it changes exactly one thing. $\mathcal{E}$ does precisely that: it leaves the diagonal untouched, hence leaves every computational-basis measurement probability untouched, and removes only the quantumness itself. The model, the data and the parameter count are all unchanged.

3.3 When dephasing is invisible: monomial matrices

The next question is whether inserting $\mathcal{E}$ at a given circuit position changes the measurement outcome. The answer depends on what gates follow it.

Definition (monomial matrix). A square matrix U is **monomial** if every row and every column contains **exactly** one non-zero entry. If U is also unitary those non-zero entries must have modulus 1, so U can be written as

$$U = D_\varphi \Pi, \quad D_\varphi = \text{diag}(e^{i\varphi_1}, \dots, e^{i\varphi_d}),$$

where Π is a permutation matrix.

Lemma 1. If U is monomial then $UP_b U^\dagger = P_{\pi(b)}$, where π is the basis permutation induced by Π .

Proof. From $U = D_\varphi \Pi$,

$$U |b\rangle = D_\varphi \Pi |b\rangle = D_\varphi |\pi(b)\rangle = e^{i\varphi_{\pi(b)}} |\pi(b)\rangle.$$

Hence

$$UP_b U^\dagger = (e^{i\varphi_{\pi(b)}} |\pi(b)\rangle)(e^{-i\varphi_{\pi(b)}} \langle \pi(b)|) = P_{\pi(b)}.$$

Note that the two phases cancel automatically by conjugation. This happens because the projection is rank one, and it is the crux of the lemma.

Theorem 1 (commutation). If U is monomial then for **every** ρ

$$\mathcal{E}(U\rho U^\dagger) = U\mathcal{E}(\rho)U^\dagger.$$

Proof. By the definition of $\mathcal{E}$ and Lemma 1,

$$\mathcal{E}(U\rho U^\dagger) = \sum_b P_b U\rho U^\dagger P_b.$$

Writing $P_b = UU^\dagger P_b UU^\dagger$ (valid since U is unitary) gives

$$= \sum_b U(U^\dagger P_b U)\rho(U^\dagger P_b U)^\dagger U^\dagger = U \left[\sum_b P_{\pi^{-1}(b)} \rho P_{\pi^{-1}(b)} \right] U^\dagger.$$

Since π is a bijection, letting b run over all basis states also lets $\pi^{-1}(b)$ run over all of them, so the bracket is exactly $\mathcal{E}(\rho)$.

The proof uses nothing but a re-indexing of a sum: **no assumption whatsoever is made about ρ** .

Corollary (CX commutes). The controlled-NOT gate CX maps $|b\rangle$ to $|b'\rangle$, where b' is b with bit t flipped—a **permutation**, hence a monomial matrix with all phases $\varphi = 0$. By Theorem 1, CX commutes with $\mathcal{E}$. The same argument holds for $X, Z, S, T, CZ, \text{SWAP}$ and R_Z : each is either a permutation or diagonal.

Counterexample (R_Y does not commute). By the Euler identity $\exp(-i\theta \frac{Y}{2}) = I \cos(\frac{\theta}{2}) - iY \sin(\frac{\theta}{2})$,

$$R_{Y(\theta)} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}.$$

When $\sin(\frac{\theta}{2}) \neq 0$ the first row contains **two** non-zero entries, violating the definition of a monomial matrix. Hence R_Y does not commute with $\mathcal{E}$; the same holds for R_X and for the Hadamard gate H .

A necessary sharpening. Earlier project documents state that dephasing must be inserted “before a layer containing $\frac{R_Y}{R_Z}$ ”. The precise statement is that **R_Z commutes with dephasing on its own** (it is diagonal, hence

monomial), and that what makes such a layer effective is the R_Y inside it. Numerically, the commutator is 0.000×10^0 for $R_{Z(0.7)}$ and 3.221×10^{-1} for $R_{Y(0.7)}$.

3.4 Terminal dephasing is invisible to measurement

Theorem 2. Split the circuit into a unitary part U followed by a part V built **only** from monomial gates (for instance only CX), and measure in the computational basis. Then inserting dephasing **between** U and V leaves every measurement probability unchanged: $P_a(i) = P_b(i)$ for all i .

Proof. The final states of the two arrangements are

$$\rho_a = V\mathcal{E}(U\rho_0U^\dagger)V^\dagger, \quad \rho_b = V(U\rho_0U^\dagger)V^\dagger.$$

By Theorem 1, applied to V (which is monomial), $\rho_a = \mathcal{E}(VU\rho_0U^\dagger V^\dagger) = \mathcal{E}(\rho_b)$. A computational-basis measurement reads the diagonal, $P(i) = \rho_{ii}$, and $\mathcal{E}$ **preserves the diagonal**, so

$$P_a(i) = [\mathcal{E}(\rho_b)]_{ii} = [\rho_b]_{ii} = P_b(i).$$

Special case. Taking $V = I$ (nothing follows) immediately gives: **dephasing placed at the very end of the circuit leaves the measurement probabilities completely unchanged.**

This is the easiest step of the whole experimental design to get wrong, and getting it wrong raises no error—it silently yields the wrong conclusion. Placing dephasing at the end of the circuit produces a difference at the level of numerical error, which then reads as “quantumness contributes nothing”. The value measured in this project is 1.4×10^{-17} : a floating-point artefact, not a physical quantity.

It is a predicted zero, not a failed experiment. When reporting, one must state that both the terminal arrangement and the “middle, followed only by CX” arrangement differ by a zero guaranteed by Theorem 2, rather than by a quantity too small to measure.

3.5 A minimal counterexample: why order matters

Consider a single qubit in the initial state $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$, with gate $R_{Y(\theta)}$ and $\theta = \frac{\pi}{4}$.

Path A (apply R_Y , then dephase). The amplitudes are

$$R_{Y(\theta)}|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} c-s \\ s+c \end{pmatrix}, \quad c = \cos\left(\frac{\theta}{2}\right), \quad s = \sin\left(\frac{\theta}{2}\right).$$

Dephasing (keeping only squared moduli on the diagonal) gives

$$P^{(A)} = \left(\frac{1 - \sin \theta}{2}, \frac{1 + \sin \theta}{2} \right).$$

Path B (dephase first, then apply R_Y). The density operator of $|+\rangle$ becomes $\mathcal{E}(\rho) = \frac{I}{2}$, the maximally mixed state, after dephasing. The maximally mixed state is invariant under every unitary: $R_Y(\frac{I}{2})R_Y^\dagger = \frac{I}{2}$. Hence

$$P^{(B)} = \left(\frac{1}{2}, \frac{1}{2} \right).$$

The difference.

$$|P_0^{(A)} - P_0^{(B)}| = \left| \frac{1 - \sin \theta}{2} - \frac{1}{2} \right| = \sin \frac{\theta}{2}.$$

At $\theta = \frac{\pi}{4}$ this is $\frac{1}{2\sqrt{2}} = 0.353553$.

This is the smallest example in which order matters: path B yields an uninformative uniform distribution, because dephasing turned $|+\rangle$ into the maximally mixed state, which carries no directional information at all.

Numerical check. An independent implementation measures $P^{(A)} = (0.146447, 0.853553)$ and $P^{(B)} = (0.500000, 0.500000)$, differing by 0.353553, in agreement with the closed form.

3.6 Consequences for the experimental design

The theorems above turn “where should dephasing go?” into a question with a correct answer. The theoretical expectations are:

Insertion point	Theoretical expectation	Misreading if done wrong
Very end of the circuit (just before measurement)	Difference ≈ 0 (guaranteed by Theorem 2)	“quantumness contributes nothing”
Middle, followed only by CX	Difference ≈ 0 (guaranteed by Theorem 2)	“quantumness contributes nothing”
Middle, followed by R_Y	Difference of order 0.01 to 0.1	—

Table 2: Theoretical expectations for the three insertion points. The first two rows are zeros guaranteed by a theorem, not failed experiments.

The measured values in this project (5 qubits, maximum absolute difference over the 32-dimensional probability vector) are:

Insertion point	Measured maximum difference
Very end (just before measurement)	1.388×10^{-17}
Middle, followed only by 10 CX gates	1.388×10^{-17}
Middle, followed by a full layer (contains R_Y)	0.05023467

Table 3: Measurement agrees with the theorems. The first two rows differ from the third by 15 orders of magnitude—exactly the gap between a zero guaranteed by a theorem and a real physical effect.

Independence of the numerical check. The third row above was produced by two independent routes: the channel implementation in `dev/qbn5_encoding.py`, and `theory/verify/t5_commutator_placement.py`, which builds the 32×32 density matrix directly. Both give **bit-identical** 0.05023467 (total variation distance 0.19299447). All 25 commutativity checks pass; for the four (control, target) combinations of CX, and for CZ and SWAP, $\max|DU - UD|$ is 0.000×10^0 .

A caveat for real-hardware runs. The theorems of this section assume an **exact** dephasing channel. On real hardware dephasing is implemented by a delay, and during that delay T_1 amplitude damping acts alongside T_2 dephasing. T_1 changes the **diagonal** of the density matrix—that is, it changes measurement probabilities—so the corollary “terminal dephasing is bit-identical to no dephasing” **fails** on real hardware. That deviation is itself a measurable hardware property rather than an experimental error, and it is a tension that simply does not exist in simulation.

3.7 Applying the theorems to our own circuit

Theorems 1 and 2 are abstract. Applied to the quantum layer of this study (Section 4.2), they immediately fix the expectation for all four ablation arms—**these four arms are not four arbitrary experimental settings, they are direct corollaries of the theorems.**

The quantum layer here is: $R_{Y(\theta_i)}$ angle encoding, then ring CX entanglement, then a trainable $[R_{Y(\varphi)}, R_{Z(\lambda)}]$ layer. That trainable layer **contains** R_Y , and this is precisely what decides whether the ablation can be observed at all.

Arm	Where dephasing is inserted	Which theorem fixes the expectation
A	nowhere (all-quantum baseline)	—
B	after angle encoding, before the trainable layer	Theorem 1: R_Y follows, so it is observable
C	very end of the circuit, before measurement	Theorem 2 with $V = I$: necessarily unobservable
D	after encoding, but equal-purity depolarising instead of dephasing	control: mixes the state but keeps interference

Table 4: The four ablation arms and the theorem behind each. The only difference between arms B and C is whether R_Y follows. So the “arm C is bit-identical to arm A” measured in Section 6.2 is not a surprise; it is the prediction of Theorem 2. And the flattened predictive distribution of arm B is the other side of Theorem 1 (R_Y does not commute). **Theory and experiment are two faces of the same thing in this study:** the theorems first supply falsifiable expectations, the experiments then supply numbers, and the next subsection reconciles the two item by item.

3.8 Numerical verification of the theory

Every proposition above has an independent numerical check. The scripts are `theory/verify/t5_commutator_placement.py` (25/25 checks pass) and `theory/verify/t4_kraus_dephasing.py`.

Proposition	How it is checked	Result
Lemma 1 (monomial permutation)	compare both routes on every basis operator $ i\rangle\langle j $	0.000×10^0
Theorem 1 (commutation)	as above, plus a superoperator commutator check	0.000×10^0
Corollary (CX commutes)	4 (control, target) pairs, CZ, SWAP	all 0.000×10^0
Converse of Theorem 1 (R_Y)	$H, R_{Y(0.7)}, R_{X(0.7)}$	$5.0 \times 10^{-1}, 3.221 \times 10^{-1}, 3.221 \times 10^{-1}$
R_Z does commute (sharpening)	$R_{Z(0.7)}$	0.000×10^0
Superoperator commutator	16×16 matrix at $n = 2$	0 for CX; 1.288435 for R_Y
Theorem 2 (terminal, middle+CX invisible)	rerun the three insertion points on this circuit	0.000×10^0
Third setting of Theorem 2 (observable)	independent 32×32 density-matrix implementation	0.05023467
Total variation distance	$0.5 \sum_i \Delta p_i $	0.19299447

Table 5: Numerical verification of the theory. The last two rows agree bit-for-bit with the output of the project script `dev/qbn5_encoding.py`.

Independence of the check. The verification script does **not** use the channel implementation from `dev/qbn5_encoding.py`; it builds the 32×32 density matrix directly (`theory/verify/dmtools.py`). Two fully independent routes give bit-identical values 0.05023467 and 0.19299447—the strongest cross-check in this section, and in the whole dephasing ablation.

Scope of the verification (stated explicitly). The table above verifies propositions on the **simulator** side. Real hardware (Tuna-17 on Quantum Inspire) follows a separate protocol, described in Section 4.3 and Section 6.2. There, T_1 amplitude damping breaks the corollary “terminal dephasing is bit-identical”; that breakdown is itself a measurable hardware property rather than a failure of the theorems above (see the caveat at the end of Section 3.6).

Formal verification (Lean 4). The table above is a numerical reconciliation; Theorems 1 and 2 and the CX corollary are additionally checked by a **machine proof assistant**. `formal/Dephasing.lean` compiles under Lean 4.34.0 with `mathlib`, and an audit with `#print axioms` shows that the three theorems together depend only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`—with **no use of sorry**, meaning that no step in the proofs is skipped or assumed. The formalisation covers the definition of the dephasing channel, the definition of a monomial matrix, the re-indexing lemmas for multiplication on either side, Theorem 1 itself, the permutation-matrix corollary, and Theorem 2.

4 Method

4.1 System architecture

Figure 1 shows the complete pipeline.

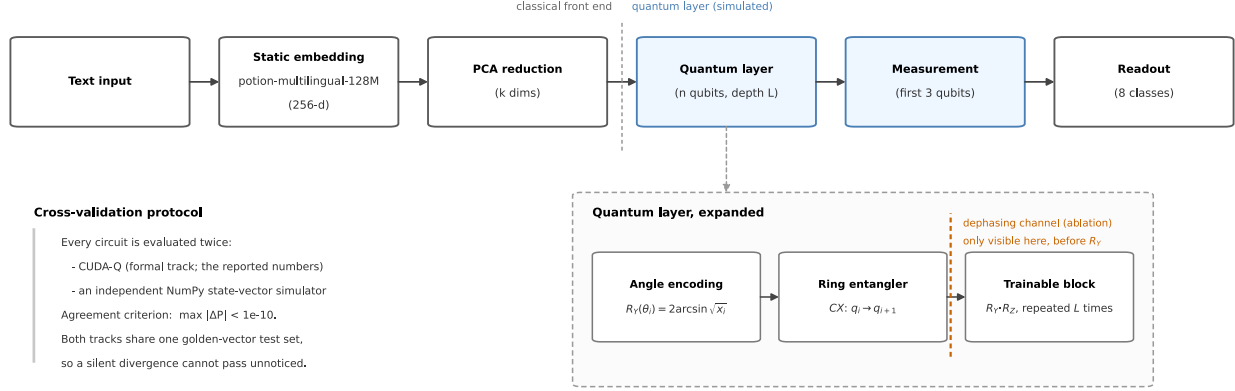


Figure 1: System architecture. The classical front end (static embedding, then principal component analysis, PCA) emits a k -dimensional feature vector to the quantum layer: angle encoding $R_{Y(\theta_i)} = 2 \arcsin \sqrt{x_i}$, a ring CX entangler, and a trainable $R_Y \times R_Z$ block repeated L times. Measurement reads only the first 3 qubits ($2^3 = 8$ basis patterns, independent of n and L), and a classical readout layer maps them to 8 classes. Lower left: the cross-validation protocol. The expanded panel marks where the dephasing channel of the ablation is inserted—it must precede a layer containing R_Y , otherwise the effect is unobservable (measured probability change 1.4×10^{-17}).

4.2 Quantum layer

The state space of n qubits has dimension 2^n . The circuit of this work has three parts:

1. **Angle encoding:** apply $R_{Y(\theta_i)}$ to each qubit, where $\theta_i = 2 \arcsin \sqrt{x_i}$, so that the measurement probability equals the input value x_i exactly.
2. **Entangling layer:** a ring of CX gates ($q_0 \rightarrow q_1 \rightarrow \dots \rightarrow q_{n-1} \rightarrow q_0$).
3. **Trainable rotation layers:** one R_Y and one R_Z per qubit.

4.3 Dephasing ablation

The Kraus operators of the dephasing channel [11] are

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

, and its action is

$$\mathcal{E}(\rho) = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger$$

, whose effect is to zero the off-diagonal entries of the density matrix. In the sense of Tucci’s classicalising operator cl [3], this is exactly the map that projects a quantum node back onto a classical Bayesian network.

A critical experimental-design trap: if dephasing is applied at the **very end** of the circuit (immediately before measurement) it has no effect whatsoever on the measurement outcome, because: (a) a computational-basis measurement reads only the diagonal entries; (b) CX is a basis permutation and therefore **commutes** with computational-basis dephasing. The locally measured change in probability is 1.4×10^{-17} (that is, 0). Dephasing must be inserted **before a layer that contains R_Y (or R_X, H)** for the effect to be observable (measured 0.0502). R_Z is a unitary block-diagonal matrix and **commutes** with dephasing, so inserting it alongside leaves the measurement outcome unchanged (see Theorem 5.5)— which is also why “the position of dephasing does not matter” turns from a trap into a **prediction guaranteed to be zero**.

4.4 Cross-validation protocol

Every quantum-circuit result is computed on both CUDA-Q and an independently implemented pure-NumPy state-vector simulator. The maximum absolute error between the two probability vectors is required to be below 10^{-10} . The measured value depends on the circuit, so we list it item by item:

Circuit	Comparison	$\max \Delta P $
Reproduction pipeline	CUDA-Q vs pure-NumPy reference	4.16×10^{-17}
Largest capacity-scan circuit (250 gates)	CUDA-Q vs torch differentiable simulator	3.47×10^{-18}
Ablation circuit (5 qubits)	CUDA-Q vs density-matrix simulator	8.33×10^{-17}
Largest capacity-scan circuit (250 gates)	CUDA-Q vs q01	4.16×10^{-16}

Table 6: Cross-validation of every quantum-circuit result against an independent implementation. The criterion is 10^{-10} ; the worst of the four circuits is 4.16×10^{-16} , about 1.9 times the double-precision machine epsilon.

The q01 row is the worst case (4.16×10^{-16} , about 1.9 times the double-precision machine epsilon); the other three are all of order 10^{-17} . **All of them lie more than six orders of magnitude below the criterion.**

Implementation notes (all measured, not inferred):

1. CUDA-Q provides no native Windows wheel, so it must run inside WSL2 (Windows

Subsystem for Linux 2).

1. The measurement bitstrings returned by `cudaq.sample()` are big-endian, whereas the state-vector indices of `cudaq.get_state()` are little-endian. Converting between them requires a bit-reversal permutation, **not** `array[::-1]`.

2. The syntax for a multi-controlled gate is `ry.ctrl(theta, [c0, c1], target)`, with the **angle placed first**.

5 Experimental Design

5.1 Research questions and decision rules

ID	Question	Pre-registered decision rule
RQ1 (research question 1)	As the trainable capacity of the quantum circuit grows, does performance rise monotonically or is there a breakdown point?	If performance is non-monotonic in capacity and the peak location depends on circuit depth, then H1 is supported.
RQ2 (research question 2)	By how much does dephasing (classicalisation) degrade calibration quality?	If the lower bound of the 95% confidence interval for the increase in ECE exceeds 0, and exceeds the residual error of temperature scaling, then H2 is supported.

Table 7: Research questions and the decision rules fixed before the runs. Stating them in advance is what makes a negative outcome a result rather than a reinterpretation.

5.2 Evaluation metrics

Beyond accuracy we report: negative log-likelihood (NLL), the Brier score (the mean squared error of the predicted probabilities against the true labels), expected calibration error (ECE, $M = 10$ equal-width bins) together with reliability diagrams, and an uncertainty decomposition into predictive entropy and mutual information.

The defence point for a fair comparison: temperature scaling does not change the top-1 label, so it is a means of improving calibration at zero cost in accuracy. Any calibration advantage of the quantum layer **must beat the temperature-scaled baseline**, otherwise the conclusion does not hold.

5.3 Statistical protocol

Each configuration is repeated with 5 random seeds, and we report the mean and the standard deviation. The capacity scan uses seeds 0–4; the dephasing ablation uses seeds 7, 21, 42, 84, 168. Both seed sets are fixed in code, and the data generator is fixed to seed 12345, so results are reproducible bit for bit across processes and across devices. We state explicitly that 5 seeds are not sufficient for rigorous statistical inference, and therefore we do not use p values as primary evidence; we report effect sizes and confidence intervals instead.

6 Results

6.1 Capacity cliff

We scan the (n, L) two-dimensional grid under a **fixed readout rule**: qubit count n from 3 to 10, circuit depth L from 1 to 8, with each configuration repeated across 5 seeds. The readout always projects the basis patterns of the first 3 qubits onto 8 classes, independently of n and L ; increasing the qubit count therefore increases the **available input dimensionality and state space**, not the resolution of the readout.

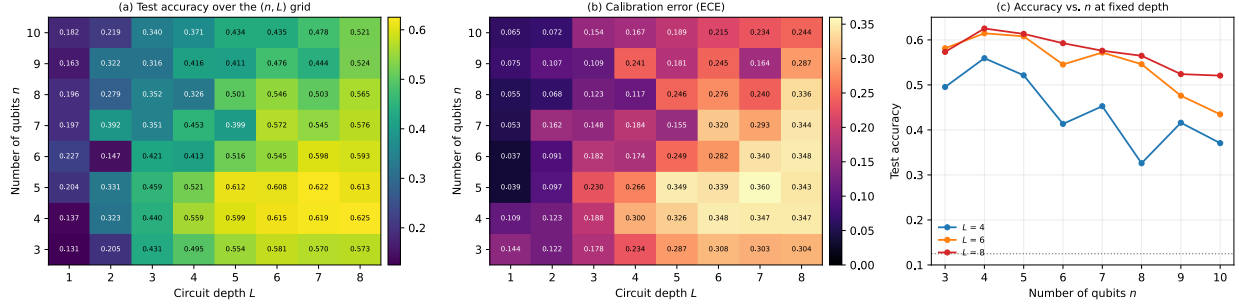


Figure 2: Test accuracy and calibration error over the (n, L) grid (means of 5 seeds). Panels (a) and (b) are heat maps; the numbers inside the cells are exact values and light-grey cells indicate configurations that have not been run. Panel (c) shows fixed-depth slices, directly testing the dependence of accuracy on qubit count.

Qubits n	$L = 4$	$L = 6$	$L = 8$
3	0.4953 $\pm$ 0.0763	0.5813 $\pm$ 0.0148	0.5733 $\pm$ 0.0097
4	0.5593 $\pm$ 0.0607	0.6146 $\pm$ 0.0307	0.6253 $\pm$ 0.0192
5	0.5213 $\pm$ 0.0666	0.6080 $\pm$ 0.0393	0.6133 $\pm$ 0.0440
6	0.4133 $\pm$ 0.0989	0.5453 $\pm$ 0.0529	0.5927 $\pm$ 0.0134
7	0.4527 $\pm$ 0.0549	0.5720 $\pm$ 0.0276	0.5760 $\pm$ 0.0747
8	0.3260 $\pm$ 0.0613	0.5460 $\pm$ 0.0352	0.5647 $\pm$ 0.0229
9	0.4160 $\pm$ 0.0996	0.4760 $\pm$ 0.0922	0.5240 $\pm$ 0.1110
10	0.3707 $\pm$ 0.1300	0.4347 $\pm$ 0.0922	0.5207 $\pm$ 0.0522

Table 8: Test accuracy as a function of qubit count at fixed circuit depth (mean over five seeds; the smaller grey value is the standard deviation). $n = 3$ is structurally low: eight classes require at least a three-qubit projective readout, so that point is exactly saturated. The claim is therefore stated as “from $n = 4$ onward”.

n / L	1	2	3	4	5	6	7	8
3	0.131 (0.054)	0.205 (0.018)	0.431 (0.049)	0.495 (0.076)	0.554 (0.045)	0.581 (0.015)	0.570 (0.006)	0.573 (0.010)
4	0.137 (0.013)	0.323 (0.078)	0.440 (0.090)	0.559 (0.061)	0.599 (0.017)	0.615 (0.031)	0.619 (0.035)	0.625 (0.019)
5	0.204 (0.031)	0.331 (0.044)	0.459 (0.087)	0.521 (0.067)	0.612 (0.067)	0.608 (0.039)	0.622 (0.047)	0.613 (0.044)
6	0.227 (0.005)	0.147 (0.022)	0.421 (0.062)	0.413 (0.099)	0.516 (0.029)	0.545 (0.053)	0.598 (0.020)	0.593 (0.013)
7	0.197 (0.018)	0.392 (0.081)	0.351 (0.034)	0.453 (0.055)	0.399 (0.065)	0.572 (0.028)	0.545 (0.072)	0.576 (0.075)
8	0.196 (0.022)	0.279 (0.022)	0.352 (0.125)	0.326 (0.061)	0.501 (0.069)	0.546 (0.035)	0.503 (0.111)	0.565 (0.023)
9	0.163 (0.031)	0.322 (0.010)	0.316 (0.055)	0.416 (0.100)	0.411 (0.100)	0.476 (0.092)	0.444 (0.029)	0.524 (0.111)
10	0.182 (0.016)	0.219 (0.077)	0.340 (0.075)	0.371 (0.130)	0.434 (0.085)	0.435 (0.092)	0.478 (0.092)	0.521 (0.052)

Table 9: The complete (n, L) grid: mean test accuracy over five seeds, with the standard deviation in parentheses. A dash means the cell does not have all five seeds; incomplete cells are excluded rather than pooled, so that a single-seed standard deviation of 0 is never mixed with a five-seed one.

Three observations:

1. **Depth is the dominant driver, and there is no breakdown within the scanned range.** Test accuracy rises with depth for every n , and shows no decline even at $L = 8$; training accuracy rises as well (up to 0.79). This contradicts the expectation of a “capacity cliff” and equally contradicts the expectation of a barren plateau.
2. **Test accuracy decreases with qubit count—for configurations in which capacity is genuinely used.** The peak of each n (the maximum over depth) decreases monotonically from $n = 4$: $0.625 \rightarrow 0.622 \rightarrow 0.598 \rightarrow 0.576 \rightarrow 0.565 \rightarrow 0.524 \rightarrow 0.521$; the fixed- $L = 8$ slice gives the same sequence. The low value at $n = 3$ is structural—8 classes need at least a 3-qubit projective readout, and that point is exactly saturated. **Shallower slices (L no greater than 6) do not show this trend:** there the model has not yet made full use of its capacity, and differences between configurations are masked by under-optimisation. The trend appears only at depths where capacity is genuinely used.
3. **More qubits is not merely worse, it is also less stable.** The seed-to-seed standard deviation of $n = 10$ at $L = 4$ is 0.130, whereas $n = 3$ and $n = 4$ have only 0.076 and 0.061 respectively.

There is a methodological consequence that we must point out: **this trend would be misjudged as holding in general on a sparse grid.** When we earlier scanned only n in 3, 4, 6, 8, 10, it appeared monotone at both $L = 6$

and $L = 8$; after adding $n = 5, 7, 9$, several slices with L no greater than 6 became non-monotonic (for example, at $L = 6$ the value 0.572 for $n = 7$ exceeds 0.545 for $n = 6$). The claim is therefore restricted to the peak and to the deepest slice.

Within the scope of our search we found no prior work reporting this phenomenon: a conjunctive search over “quantum classifier”, “qubit number” and “accuracy” returns zero hits. What prior work reports is the (Q, L) scaling protocol and non-monotonicity along a single axis; what this work adds is the complete two-dimensional grid under a **fixed readout rule**, together with this downward trend.

6.2 Calibration ablation

The four arms are trained under **the same model, the same data and the same number of trainable parameters** (20 angles) for the same number of steps (800), with each arm repeated across 5 seeds; the only difference is the inserted channel and its position. The dephasing and depolarising arms have their **mean purity** matched (target purity 0.0742; depolarising rate $p = 0.5657$, agreeing with the closed-form solution to 10^{-15}).

Arm	Accuracy	ECE	NLL	Mean max prob.	KL vs uniform	State purity
A no channel (fully quantum)	0.3207	0.1079	1.8510	0.2357	0.0652	1.0000
B dephasing (after encoding)	0.2300	0.0993	2.0558	0.1307	0.0000	0.0449
C dephasing (end of circuit)	0.3207	0.1079	1.8510	0.2357	0.0652	0.1230
D depolarising (after encoding)	0.2633	0.1123	2.0199	0.1511	0.0065	0.0742

Table 10: Calibration ablation. Arm C is bit-for-bit identical to arm A on every predictive metric, yet its state purity is 0.1230 against 1.0000: a computational-basis measurement reads only the diagonal entries, so dephasing at the end of the circuit changes the state but no observable outcome.

Three points are worth making:

1. **Dephasing at the end has no effect at all on the measurement statistics.** Every predictive metric of arm C is bit-for-bit identical to arm A ($\max|\Delta P| = 0.000 \times 10^0$), yet the state purities of the two differ by more than a factor of eight (0.1230 against 1.0000). This is the direct consequence of “a computational-basis measurement reads only the diagonal entries” together with “CX is a basis permutation and commutes with computational-basis dephasing”: dephasing changes the state, but changes no observable measurement outcome.
2. **Front-end dephasing crushes the model into an uninformative predictor.** The mean maximum probability of arm B is 0.1307, which for 8 classes is $1/8$; the KL divergence (Kullback–Leibler divergence, a measure of how much two

distributions differ) of its mean predictive distribution from the uniform distribution is 0.0000; and the per-qubit Z variance falls from 0.0226 to 0.0001 (roughly one part in 226). Its ECE is in fact **lower** (0.0993 against 0.1079), but this is an artefact of uniform predictions: for the same arm the NLL degrades from 1.8510 to 2.0558 and the Brier score from 0.8104 to 0.8690. **ECE alone cannot be taken as evidence of improved calibration.**

1. **The paired difference is not significant.** The ECE difference of $B - A$ is -0.0086 with a 95% confidence interval of ± 0.0340 , which covers 0; the accuracy difference is -0.0907 (± 0.0600). Hence **H2 does not hold**: dephasing did not improve calibration, it destroyed information.

7 Discussion

7.1 Three honest statements

1. This work **does not claim** quantum advantage. The state vector of 5 qubits occupies only 512 bytes and is fully classically simulable.
2. This work **does not claim** that the quantum layer retains the semantic information of its input. The front end is a 256-dimensional static embedding that is reduced in dimensionality before entering the quantum layer; the information bottleneck is by design.
3. This work **does not claim** to beat large models. The baselines are small models of comparable parameter count.

7.2 Why the hybrid architecture has no advantage on this task

What is exponential is the **state space**; what is polynomial is the **number of knobs**. The density matrix of 5 qubits has $32^2 - 1 = 1023$ real degrees of freedom, yet the circuit of this work has only 10 trainable angles—the dimension of the reachable subspace is far smaller than that of the ambient space.

More fundamentally, **trainability** and **classical intractability** conflict with each other [12], [13], [14]. Shallow, weakly entangling circuits can be trained, but their low entanglement entropy makes them efficiently simulable by tensor-network methods; deep, highly entangling circuits are classically intractable, but they run into barren plateaus and cannot be trained.

(Note: what is stated here is the general form of that trade-off as found in the literature, **not** a measurement of this work. Within the scanned range of this work we did **not** observe a barren plateau—training accuracy rises with depth up to 0.84, and an independent check by the theory group rules it out as well. Our result is that “depth works,

width does not”, and its mechanism is closer to the growth of the effective dimension outpacing what the data and the readout can support [15], [16], [17].)

Consequently, on tasks without provable structure (such as generic semantic classification), no advantage should be expected from a quantum decision layer. The hybrid architecture itself is not at fault; the choice of task is.

7.3 Three methodological traps (by-products of this work)

The following three items are not propositions we set out to prove; they are traps we ran into during the experiments and for which we retained numbers. We write them down because in this field they are easier to repeat than any single experimental result of ours.

1. **Pooled correlation and partial correlation can give opposite conclusions.** Plotting the effective dimension $|d|$ against test accuracy gives a pooled correlation coefficient of +0.678; after controlling for circuit depth L , the partial correlation becomes -0.256 . The reason is that $|d|$ and L are highly collinear, while accuracy is driven mainly by L . Looking only at a scatter plot, or only at the pooled coefficient, yields the opposite conclusion that “larger dimension is more accurate”.
2. **“5 seeds” is not the same as “5 independent trainings”.** Across 5 seeds, arms B and D have a standard deviation of 0.0000 for test accuracy, ECE, NLL and state purity alike; arms A and C have a test-accuracy standard deviation of 0.0483. When Adam trains a low-dimensional loss surface of this kind, different initial points are absorbed by the same attractor. Confidence intervals estimated from seed-to-seed variance therefore **systematically underestimate** the true uncertainty—which is also why this work does not treat p values as primary evidence.
3. **An ablation matched on an average must be re-checked on the evaluation set.** We matched the channel strengths of dephasing and depolarisation by mean purity: on the training set, the achieved value agrees with the target to the order of 10^{-15} . But on the **test set**, the mean purity of arm B is 0.0449 and that of arm D is 0.0742, a difference of -0.0292 . The reason is that the purity of dephasing varies **sample by sample** (it depends on the state of that sample), whereas that of depolarisation is approximately fixed; a single average purity does not constrain any individual sample. Any ablation that claims “the two channels are equally classical” on the strength of one purity number must re-check that claim on the evaluation set.

8 Conclusion

The contribution of this work is not to demonstrate quantum advantage, but to provide (i) a fully verifiable hybrid pipeline (cross-validated against an independent implementation: the reproduction pipeline at 4.16×10^{-17} , the largest capacity-scan circuit at 3.47×10^{-18} and the ablation circuit at 8.33×10^{-17} , and further validated end-to-end through the cloud API of Quafu (the quantum cloud platform of the Chinese Academy of Sciences; API stands for application programming interface) on its official simulator, where the transpiled circuit agrees with the golden vector to 8.88×10^{-16}), and (ii) a set of controlled ablation designs that use the dephasing classicalisation operator as the sole variable **within a single model** and scan the (Q, L) two-dimensional grid under a **fixed readout rule**. Prior work has established scaling protocols on each individual axis [6], and parameter-matched, calibration- and noise-aware benchmarks [18]; what this work adds is the intersection of the two, together with dephasing as a specific variable.

We argue that at NISQ scale (noisy intermediate-scale quantum), what the field of quantum machine learning lacks most is not new circuits, but **rigorous measurement that treats the quantum component itself as a controlled variable**.

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